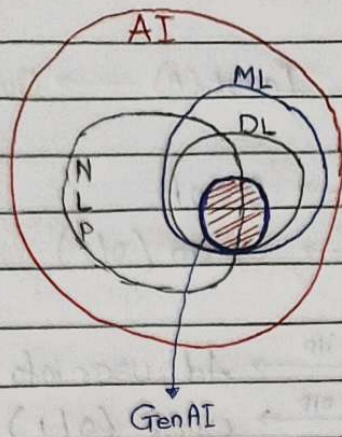


Gen AI

Date: / /



GenAI ?

A type of AI that creates new content such as text, music, images, or code by learning patterns from existing data & is capable of mimicking human creativity.

- Impacts 10% of the tasks carried out by 80% of workers and more than 50% of the tasks carried out by 20%.

→ Job loss due to automation

→ Productivity gain

* GenAI impact areas →

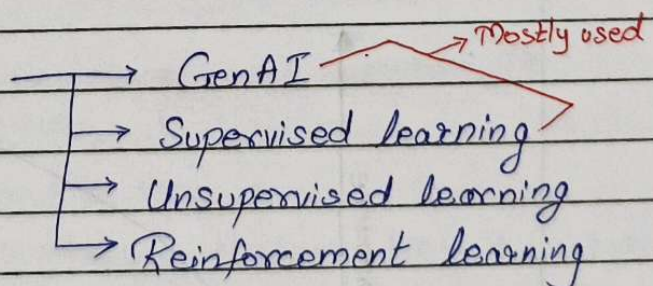
- Customer support
- Content creation
- Education
- Automation
- Software development

* If GenAI is going to be relevant ?

- 1) Does it solve real world problems?
- 2) Is it useful on daily basis?
- 3) Is it impacting the world economy?
- 4) Is it creating new jobs?

How GenAI works?

AI is a set of tools



Date: / /

* **Supervised learning** $\Rightarrow$ Input (A) $\rightarrow$ Output (B)

$\rightarrow$ Applications

$\rightarrow$ Spam filtering $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Email} \\ \text{o/p} \rightarrow \text{Spam (0/1)} \end{array} \right.$

$\rightarrow$ Online advertising $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Ad, user info} \\ \text{o/p} \rightarrow \text{click (0/1)} \end{array} \right.$

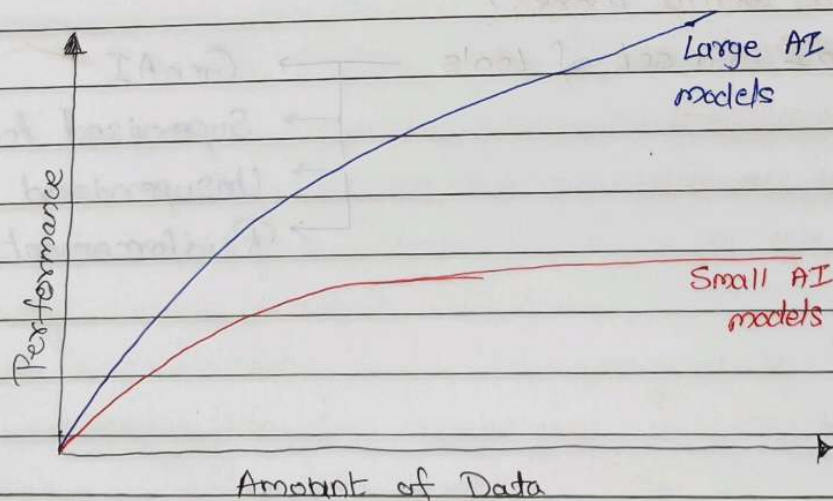
$\rightarrow$ ~~Image~~ Self-driving cars $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Image, radar info} \\ \text{o/p} \rightarrow \text{Position of other cars} \end{array} \right.$

$\rightarrow$ X-ray / Healthcare $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{X-ray image} \\ \text{o/p} \rightarrow \text{Diagnosis} \end{array} \right.$

$\rightarrow$ Visual inspection $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Image of phone} \\ \text{o/p} \rightarrow \text{Defect? (0/1)} \end{array} \right.$

$\rightarrow$ Speech recognition $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Audio recording} \\ \text{o/p} \rightarrow \text{Text transcript} \end{array} \right.$

$\rightarrow$ Reputation monitoring
OR Sentiment analysis $\left\{ \begin{array}{l} \text{i/p} \rightarrow \text{Restaurant reviews} \\ \text{o/p} \rightarrow \text{Sentiment (pos/neg)} \end{array} \right.$



Date: / /

* LLMs

- Built using supervised learning ($A \rightarrow B$)
- For text generation, they repeatedly predict the next word.

ex.

- Suppose, somewhere on internet it read
"My favourite food is a bagel with cream cheese"
- This line will be turned into a lot of datapoints for it to try and predict the next word.

Input (A)	Output (B)
My favourite food is a	bagel
" " " " " bagel	with
" " " " " with cream	cheese

- When we train a very large AI system on a lot of data (100s of billions/trillions of data words) we get a Large Language Model like chatgpt.

LLMs as a thought partner

- Can give good results with proper context.

* AI is a general purpose technology

⇒ Tasks LLMs can perform

- Writing
- Reading
- Chatting

→ web based LLM's → Chatgpt, Bard, or Bing

→ App based LLM's → email routing, Document search.

GenAI applications

* Writing

- Brainstorming product names
- Developing sales strategy
- Writing a press release
- Translation.

→ Writing a press release

Prompt → Write a press release about hiring of a new COO in our company.

⇒ This will give a general template and generic response.

Updating prompt →

If we give a more precise prompt with context about the company & the COO background, we will get more personalised output.

- Hence, with prompt we can control the o/p sounds.
- We can provide an initial prompt, review the o/p, and then ask the LLM to refine it.

* Reading

- LLMs can process, analyze, and extract value from existing text and acts as a highly efficient assistant to consume data/information.

→ Proof reading

- We can use LLM's for grammar & spell check. Also, even to fix it.

→ Summarize large articles → ex. transcription of call & summary.

→ Routing (categorization)

- Suppose, a company receives 1000s of customer review/emails.
- AI read a complaint regarding billing error, will enroute it to the finance department
- For tech bug, enroute to Tech support

→ Monitoring (Sentiment analysis)

- Beyond understanding what is being said in reviews/emails, AI can also read how it is being said.
- It can classify the emotional tone of the text (e.g. +ve, -ve, neutral, angry, frustrated, happy, etc.)

* Chatting

- This is an interactive, multi-turn conversation using LLMs.
- LLM's here ~~remembers~~ remembers what was discussed earlier in the thread.
- Entire conversation history is continuous feed-back to the model, allowing it to maintain the thread, refer back to previous points, and can answer follow-up questions.
- We can debate ideas, ask AI to role play, or ask for suggestions.

* What LLMs can do & it can't do

- LLM's have a vast field of knowledge & can speak intelligently on any topic, synthesize info. and follow instructions.
- However, like a fresh graduate, it lacks real-world, practical experience. Don't understand the physical world, and have no
- "Common sense" outside of text patterns.

* Limitations of LLM's

i) Knowledge cutoffs

- An LLM's knowledge of world is frozen at the time of its training

ii) Hallucinations

- Situation where model generate confident sounding but incorrect, misleading, or completely fabricated ~~data~~ information.

- How it happens ??

→ We already know that LLMs do not query any structured database of verified facts (like wikipedia). Instead, they predict the next "token" (word or part of a word) in a sequence based on statistical probabilities learned during training.

a) Probabilistic guessing →

- Model priority is fluency and coherence over factual truth.
- If a specific word is statistically probable based on its training data, the model will generate it, even if it is incorrect.
- "Stochastic parrot" → mimic language pattern without understanding the underlying meaning.

b) Source Amnesia →

- During training, models compress terabytes of data into mathematical weights and hence, they lose link to specific sources.
- This leads to confabulation. (a memory error, creates convincing, but fake, narratives)

Date: / /

- Causes

- Training data quality
- Overfitting / Underfitting
- Prompt ambiguity → vague, ill-structured, or misleading prompt.
- Context window Limits

- iii) Input and Output length limitations

- Can accept prompt of up to only a few thousand words.

- iv) GenAI doesn't work well with structured (tabular) data

- v) Bias and Toxicity

- Because LLM's are trained on billions of words scraped from the public internet, they absorb the biases, stereotypes, and toxic behaviour present in the data

- * **Prompting** :->

- way to give instruction to an LLM so as to guide its response.

- * Be detailed and specific

- * Guide the model to think through its answer

- * Experiment and iterate

2 practice codes in Week 2

Date: / /

Using GenAI in Software applications

* Workflow using GenAI

① Supervised Learning

Get labeled data $\Rightarrow$ Train AI model $\Rightarrow$ Deploy or run model
1 month 3 months 3 months

② Prompt-based AI

Specify Prompt $\Rightarrow$ Deploy or run model
mins/hours hours/days

ex.

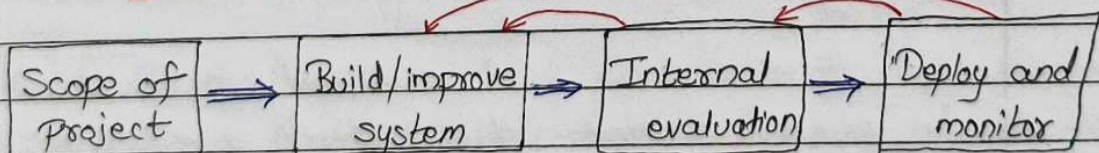
Prompt-based development

prompt = "" Classify the following review as positive or negative" : instruction

The banana pudding was really tasty! "" : Review

response = llm-response(prompt) $\rightarrow$ Code to call LLM
print(response)

Life cycle of GenAI project



What we want the software to do?

Initially a prototype, that will improve over time.

If we observe the error, we ^{move} to step 2 to improve the system

If we see mistakes, we reevaluate and send it for improvement of system.

Step 1 / Retrieval

Step 0 Preparation (Vector databases)

First, we take our private docs, chop them up into small paragraphs, and store them in a searchable system called a Vector Database.

Step 1 Retrieval

When a user asks a question, the software intercepts it. Even before AI sees the question, the software searches our Vector Database and finds the paragraph or factual document related to the query.

ex.

Query → "What is our parental leave policy?"

Action → Before AI sees the query, the system searches company's internal policy/servers and **retrieves** the exact paragraphs from HR handbook that mentions parental leave.

Step 2 Augmentation

The software takes the original query and combines (augments) it with the retrieved documents to get one massive set of hidden instructions for AI.

ex.

Action → The software automatically pastes the HR handbook text next to the original query, resulting in modified hidden prompt.

Hidden prompt → "Read the following company policy: **[inserts retrieved info]**. Now, using only this text, answer the query. "What is our parental leave policy?"

Date: / /

* Tools to improve performance

- Prompting
- Retrieval augmented generation (RAG)
 - Give LLM access to external data sources.
- Fine-tune models
 - Adapt LLM to our task
- Pretrain models
 - Train LLM from ~~at~~ scratch

Cost intuition

- 1 token $\rightarrow$ the, example, Andrew $\Rightarrow$ General words
- 2 tokens $\rightarrow$ translate, programming $\Rightarrow$ less general
- 3/4 tokens $\rightarrow$ tonkatsu

So, 1 token $\approx \frac{3}{4}$ words

Retrieval Augmented Generation (RAG)

We already know about 2 of the critical limitations of an LLM's

- Knowledge cutoff (amnesia) $\Rightarrow$ Information limitation
- Hallucination $\Rightarrow$ if doesn't know the answer, it creates a fake one.

So, this is kind of ~~an~~ a closed book test, where student relies on their memory for answers.

RAG $\Rightarrow$ A software architecture that changes this into an open-book test.

It gives the AI a temporary memory boost by letting it "read" ~~pers~~ private docs before answering.

Step 3 Generation

The LLM receives the combined hidden prompt, reads the provided references, and generates the perfectly accurate answer restricted by those facts.

Fine-tuning

The process of taking a general-purpose 'AI brain' (that already understands grammar, language, and general facts) and running an extra round of training on a small, specialized dataset to permanently change its style, tone, format, or skill set.

Why fine-tuning?

- To gain LLM with specific knowledge
- To get a smaller model to perform a task.

RAG $\Rightarrow$ Giving the AI new knowledge

Fine-tuning $\Rightarrow$ Giving the AI new behaviour

Businesses often combine both of these techniques
ex.

Fine-tuning a model so it speaks like a professional financial ~~being~~ analyst, and then wrap it in RAG system so it can search today's live stock market data to answer questions accurately.

Instruction tuning and RLHF

- Pretraining creates a 'Base model'. It is incredibly smart, but doesn't know that it is suppose to be an assistant.
- It only knows to predict the next logical word based on internet patterns.

ex.

prompt : "write a poem about a cat."

It might reply : "write a poem about a bird."

Currently, it doesn't know you were giving it an order, it thought you were just starting a list.

• To overcome this, we use "instruction tuning" and "RLHF"

i) Instruction tuning : (Teaching AI to take order.)

• AI is fed thousands of ex. of humans giving commands, paired with the correct, obedient response.

ii) Reinforcement Learning from Human Feedback :

(Teaching manners and safety)

• Helps model to become Honest, Helpful, and Harmless (HHH)
• Even if AI follows instructions, it might still be rude, dangerous, or unhelpful.

• RLHF is the final layer of training where human graders give the AI a "score" to teach it human values.

Step 1 : Train an answer quality (reward) model

ex. "Advise me on how to apply for a job"

<u>Response</u>	<u>Score (reward)</u>
Sure, here are the steps to follow	5
Just try ur best	3
It's hopeless - why bother?	1

Step 2 : Have LLM generate a lot of answers. Further, train it to generate more responses that get high scores.

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* How to choose LLMs $\Rightarrow$

mathematical connections inside its neural n/w

- Accessing model size (Parameters)

- A 1-billion parameter model is great at basic pattern recognition
- A 10-billion parameter model is good for following instructions as a chatbot
- A 100-billion parameter model can perform complex, nuanced reasoning.

- Choosing closed-source vs. open-source

- Balancing costing and latency

* Tool use and Agents

- Tool:

- External software funcⁿ connected to the AI.
- Instead of guessing, AI is programmed to recognize its own limitations and write a command to trigger the external tool.

- Agent:

- It is an LLM that is given a high-level goal, access to multiple tools, and the freedom to "think" about how to solve the problem step-by-step.

ex. "Schedule a meeting with Sarah for next Tuesday and send her agenda"

- Here AI uses calendar, contacts, email tools to carry out his multi-step workflow on its own, without need of prompt for each step.

* Augmentation vs Automation

Augmentation -

- It helps humans with a task
- It speeds up the human's work, where humans monitor or reviews everything.

ex → Recommend a response for a customer service agent to edit/approve.

Automation -

- Automatically performs the tasks entirely on its own without human oversight.

e.g. → Transcribing and summarizing records of customer interaction automatically.

Evaluating AI potential

- The potential of augmentation/automation of a task depends on:

i) Technical feasibility → can AI do it?

ii) Business value → How much value it creates?

→ faster?
→ cheaper?
→ creates value?

Artificial General Intelligence (AGI)

- An AI that can do any intellectual task that a human can